October 28, 2025

Retrieval-Augmented Generation

Outline

Background

Challenges & Possible improvement ideas

Hands-on with RAG

References:
UCB EECS CS 288. Natural Language Processing
Sharma, C. "Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers." in arXiv preprint arXiv:2506.00054, 2025.

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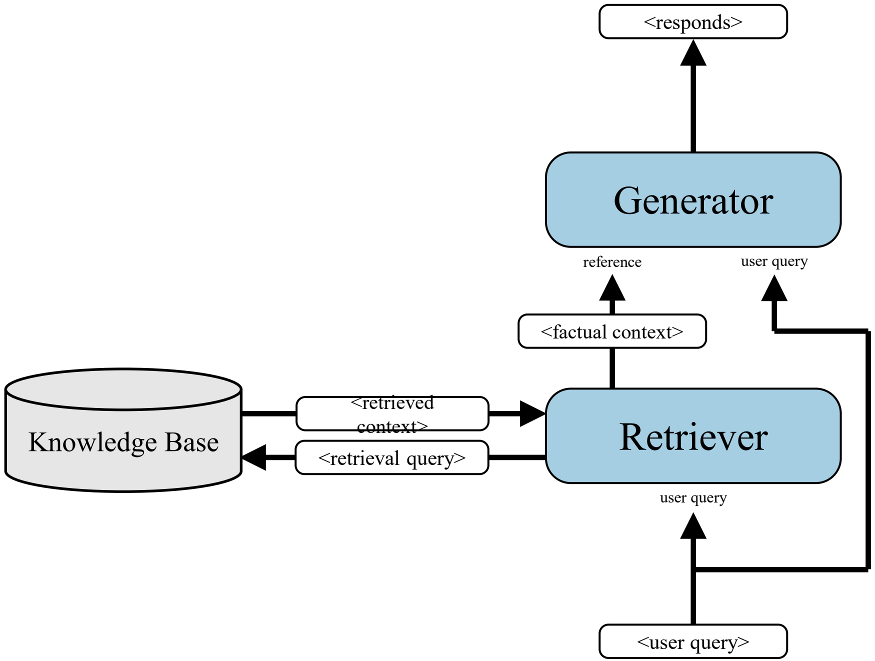

<responds>
Generator
reference
user query
<factual context>
Knowledge Base
<retrieved context>
<retrieval query>
Retriever
user query
<user query>

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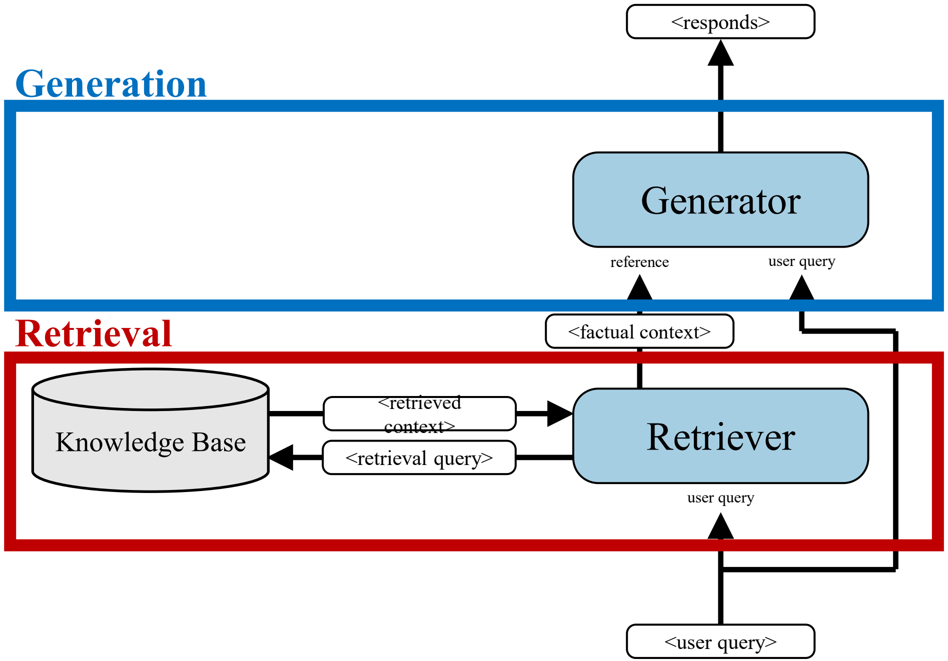

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Challenges & Possible improvement ideas – (Question Answering in specific domain)

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Challenges

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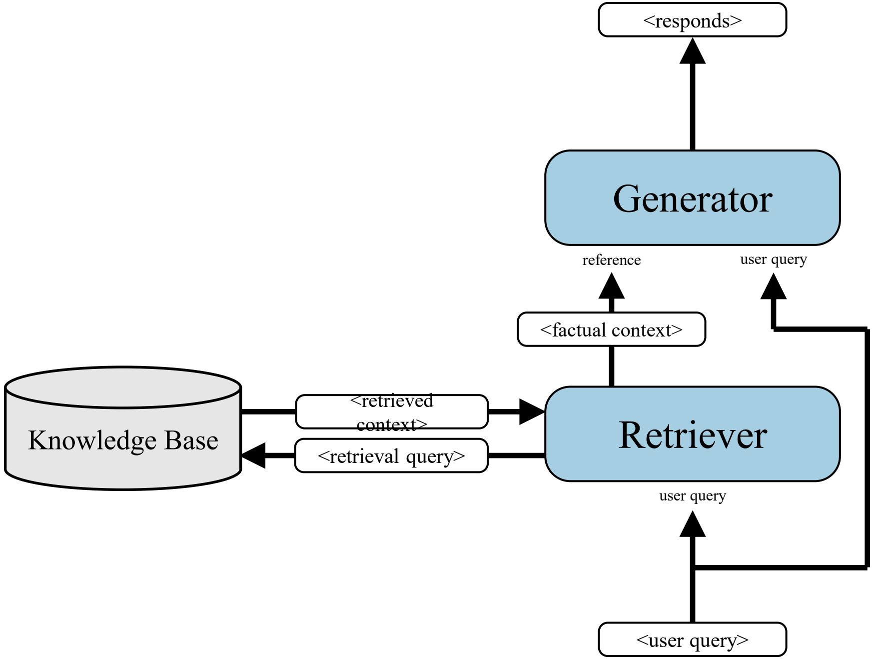

Challenges & Possible improvement ideas – (Question Answering in specific domain)

Challenges

- Retrieval quality
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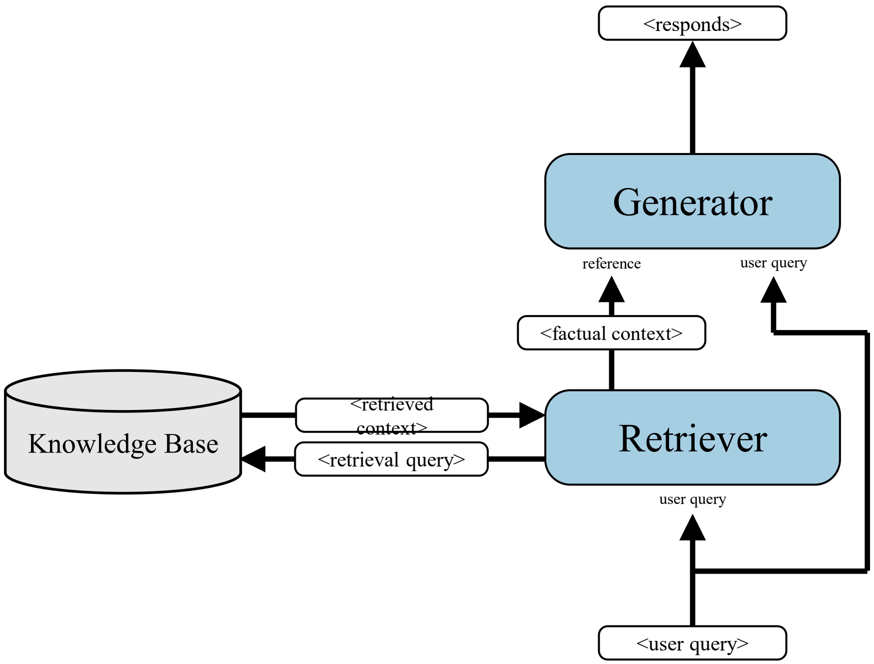

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Challenges

- Retrieval quality, Grounding fidelity
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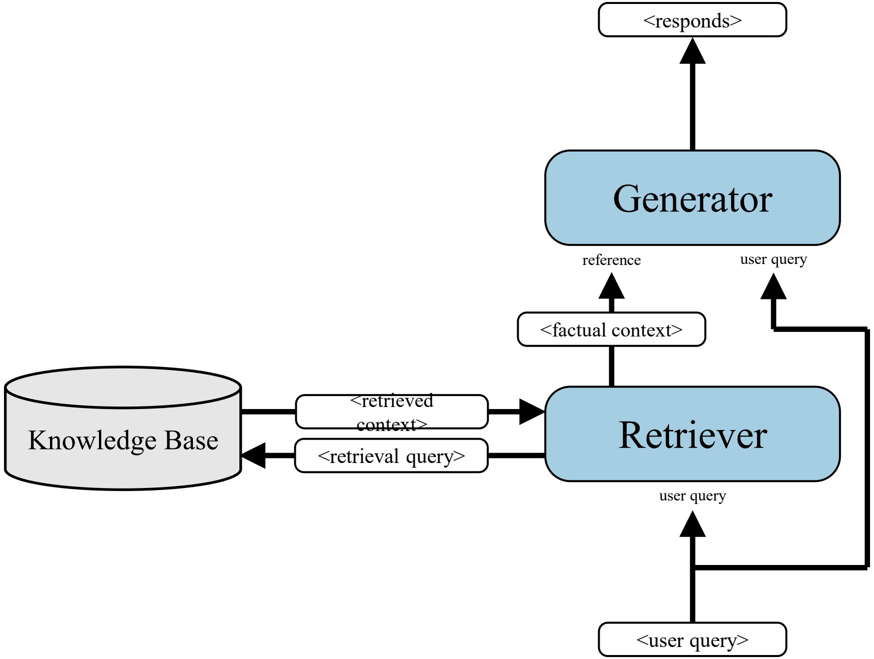

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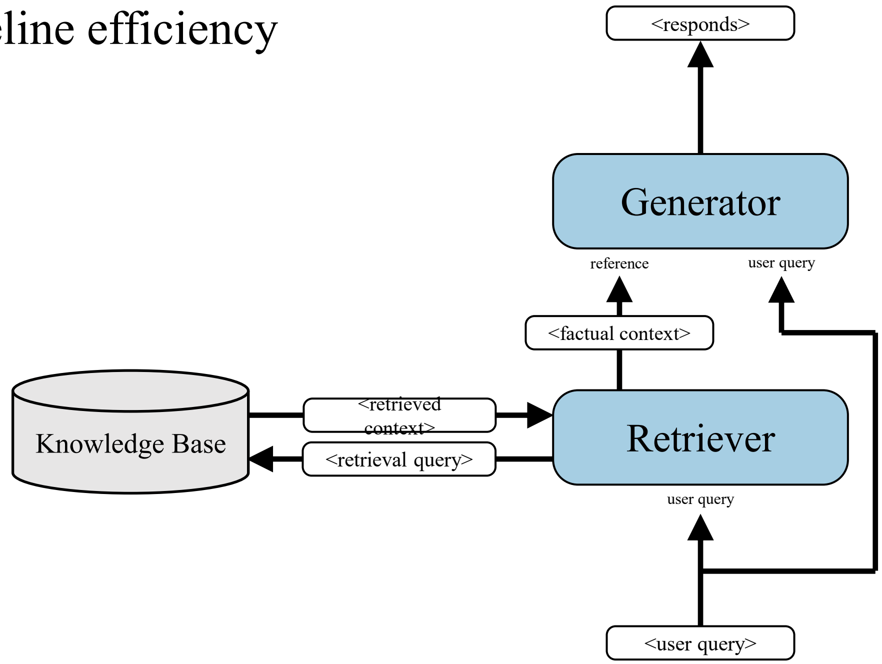

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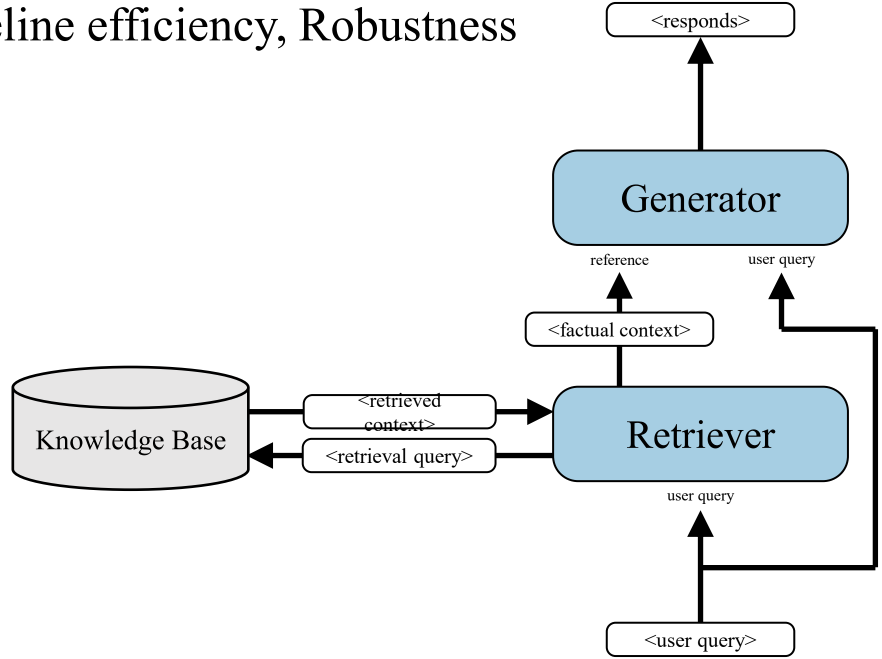

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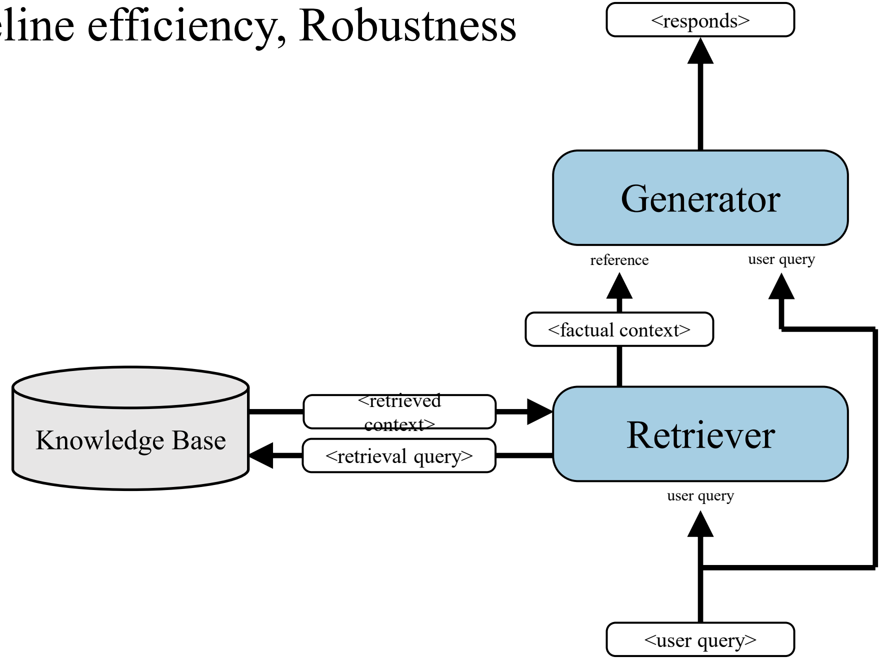
Possible improvement ideas



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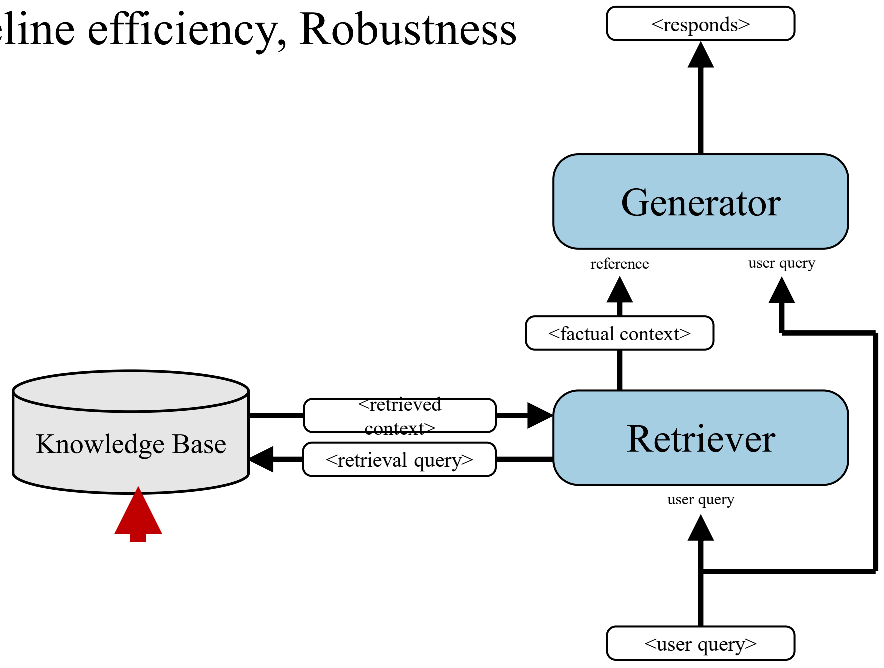

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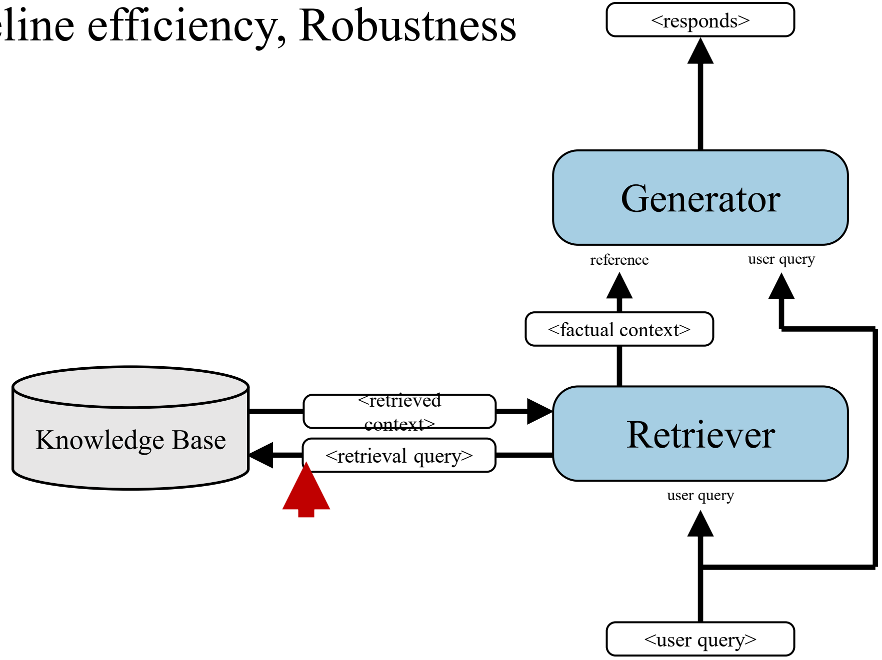

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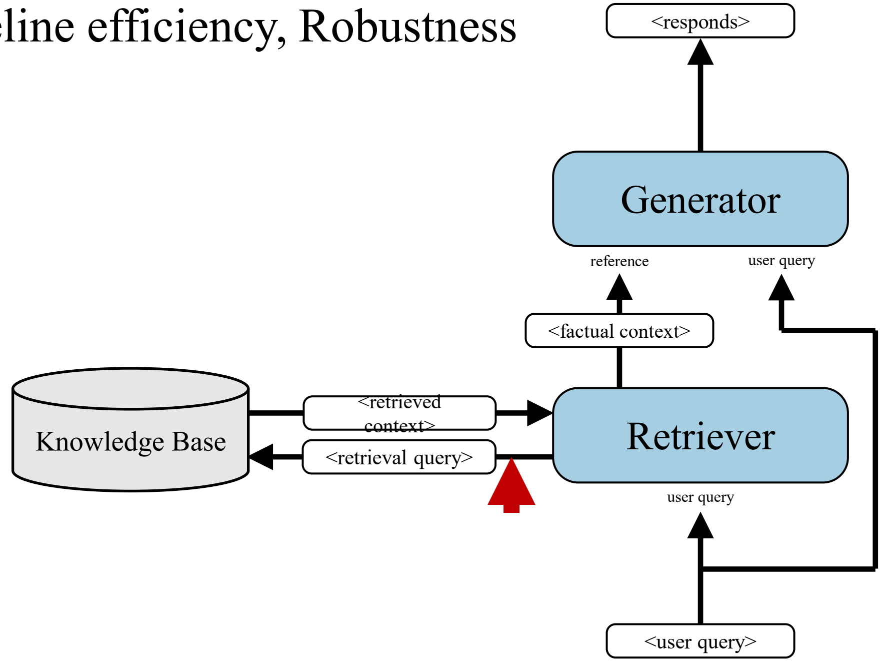

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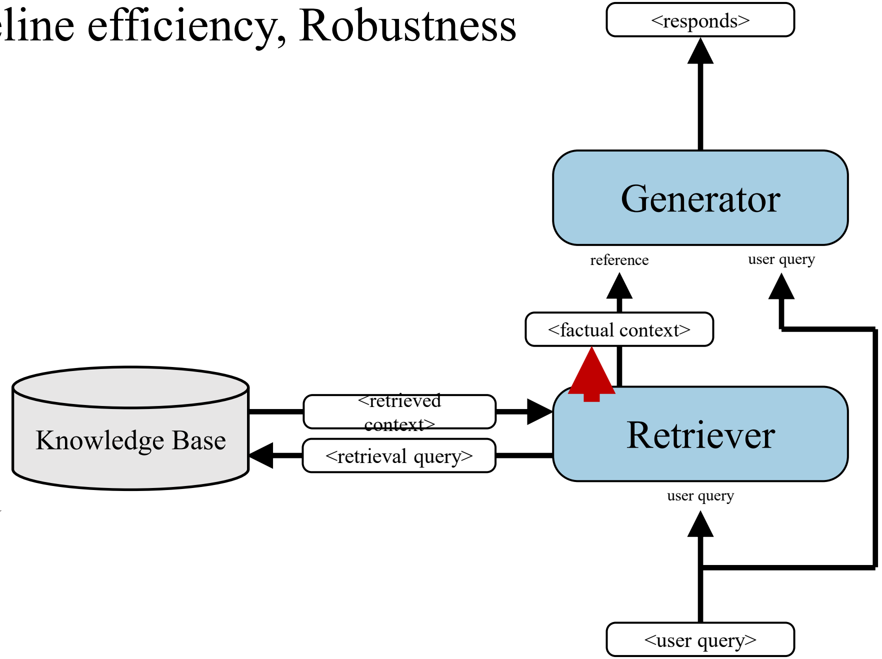

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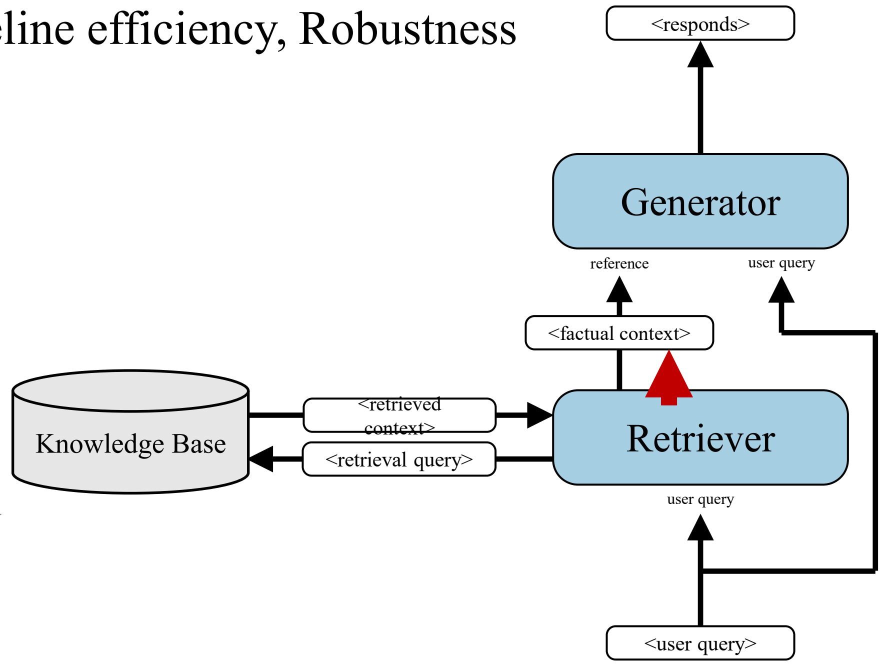

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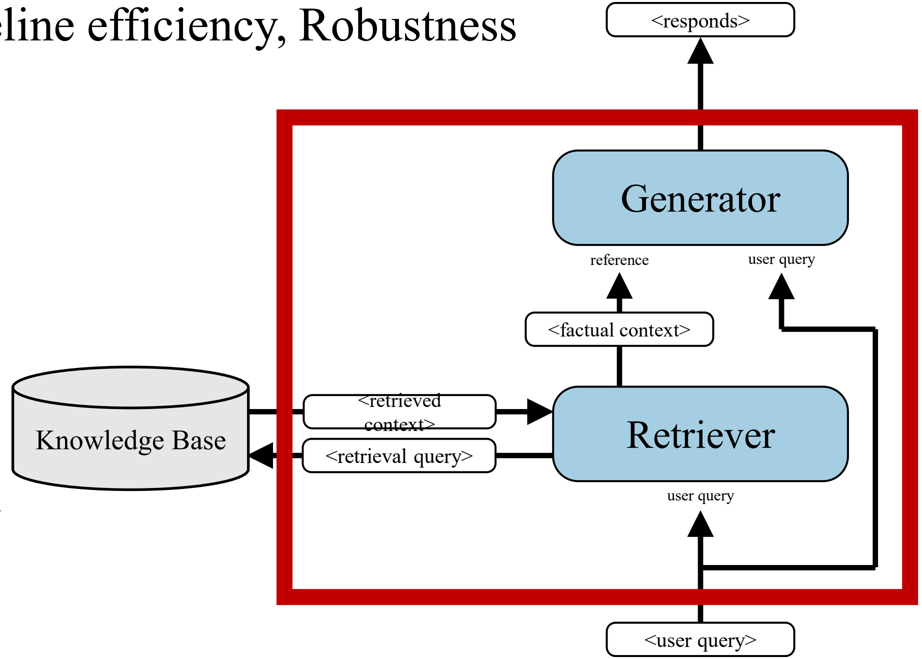
- Reranking/adaptive adoption
  use only relevant retrieved passages
- Pipeline & Framework design
  Deciding when to use retrieval? Multiple rounds of retrieval?



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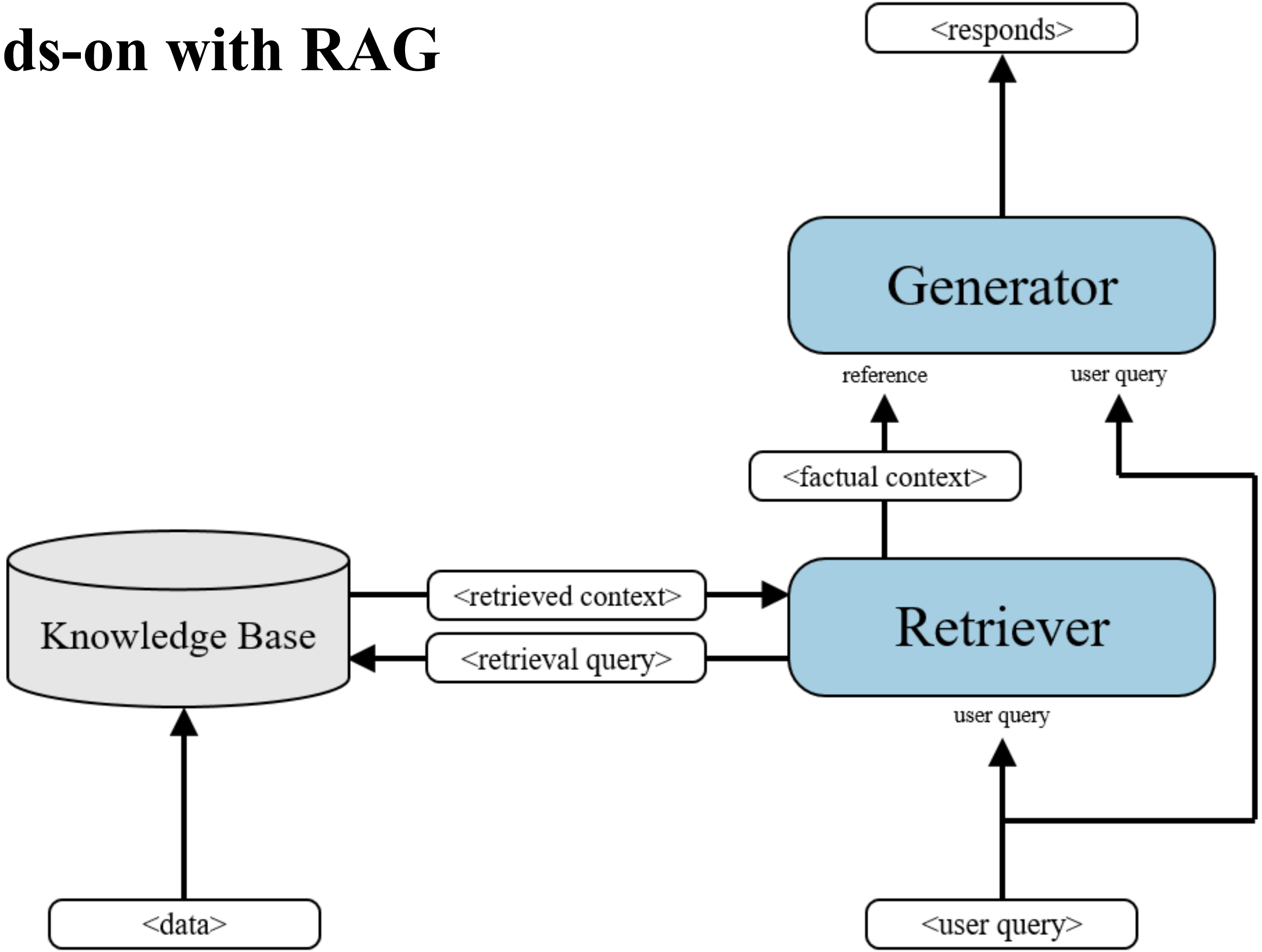

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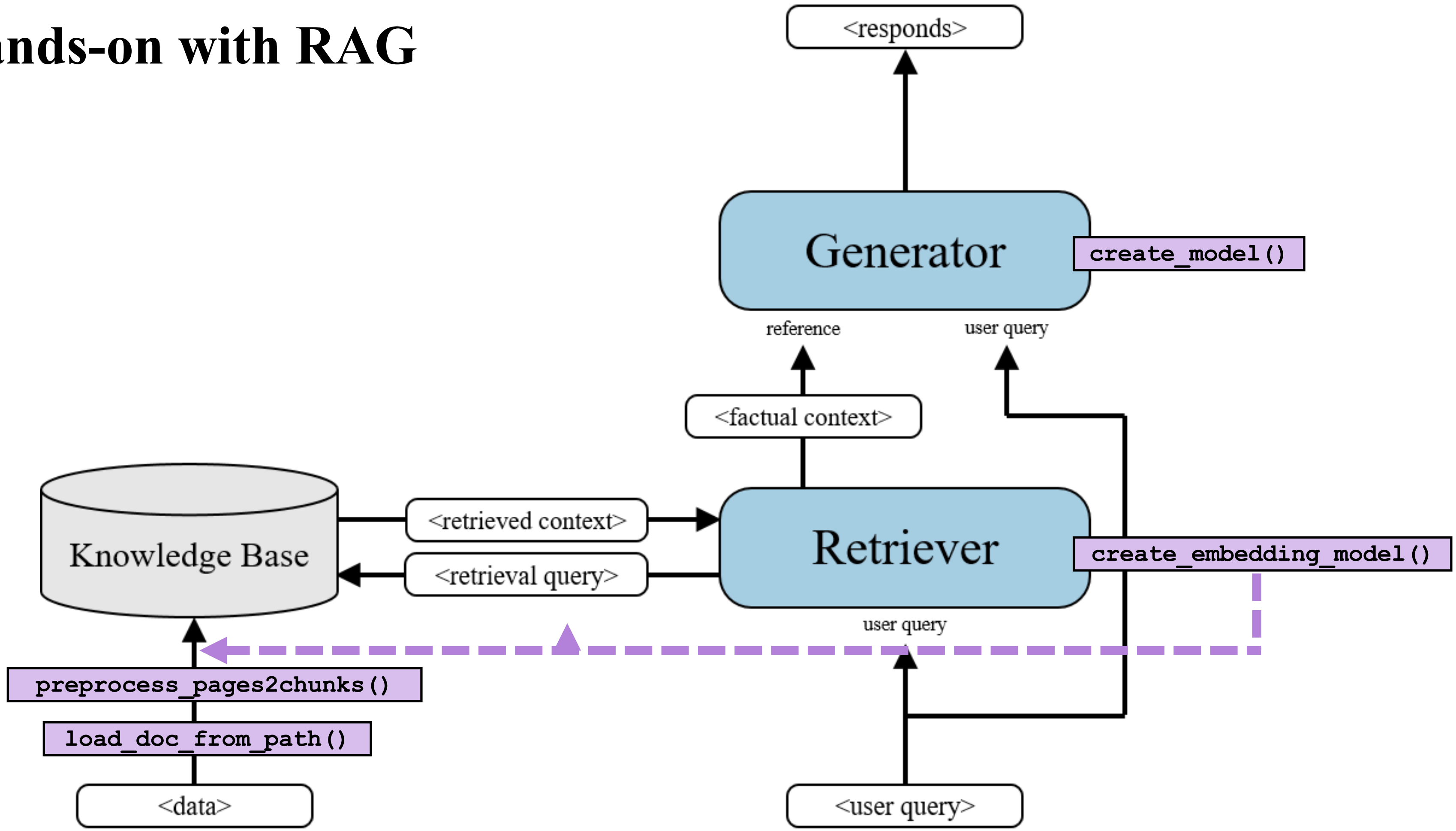

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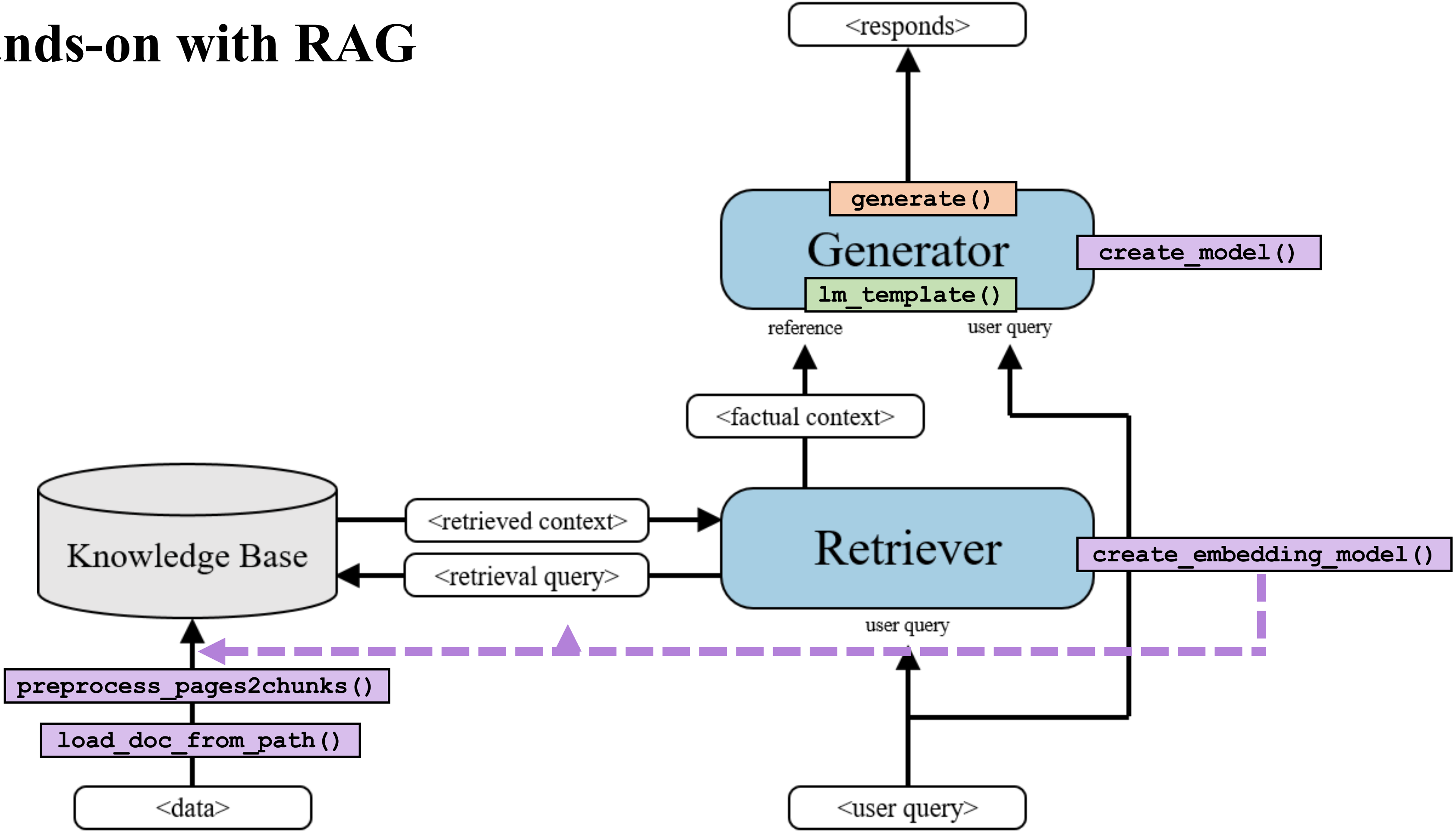

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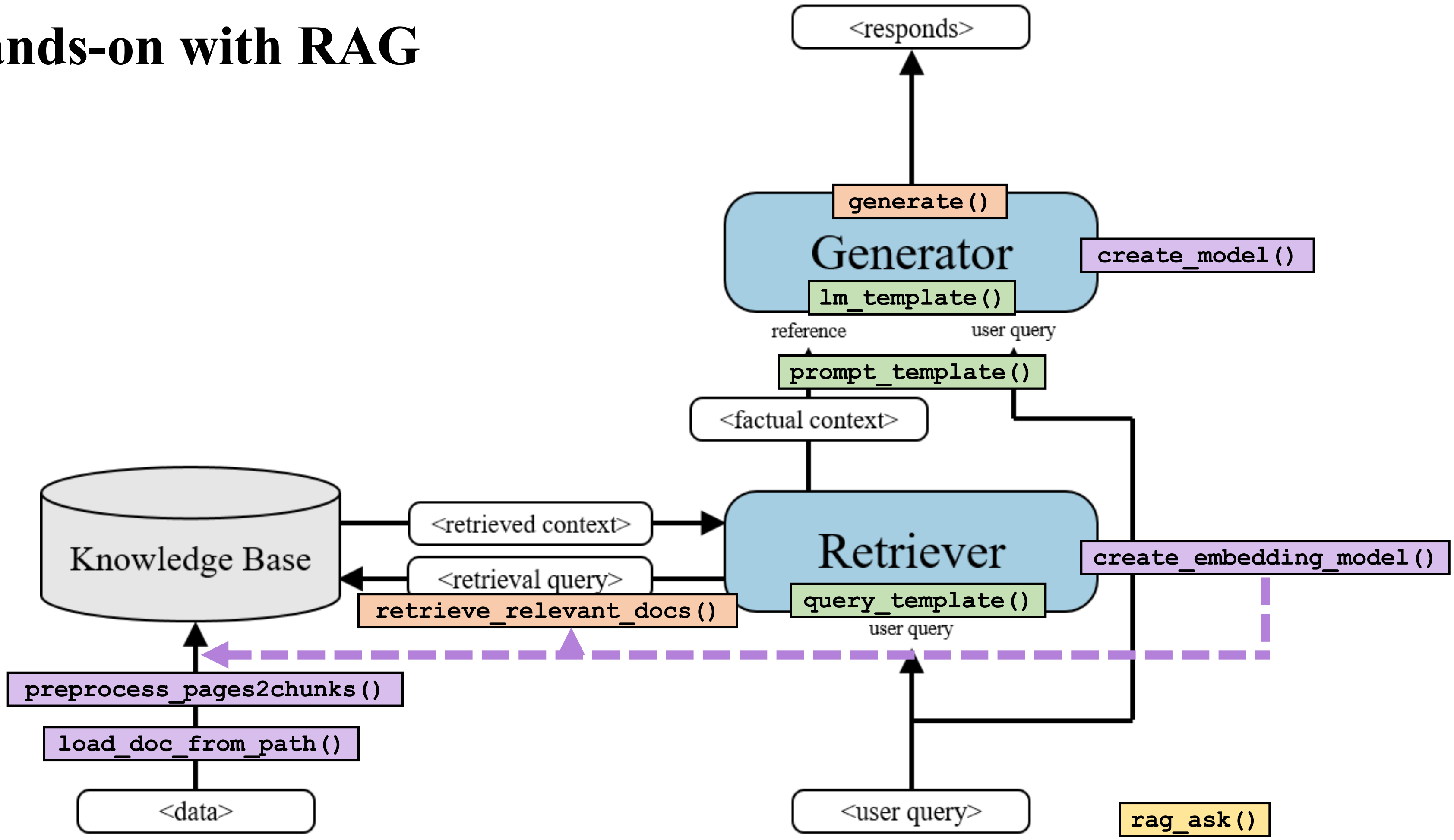


HW3 LLM

Requirements

1. **At least compare result with and without RAG**

   - Must have **evaluation metrics**: BLEU, ROUGE, BERTScore...

   - Prepare your own dataset

2. Bonus: compare different models

3. Bonus: try different RAG pipeline

Submit and Grading

1. Jupyter notebook / code
2. Report

    - Describe your dataset - 20% (notebook example)

    - Result without RAG - 20%

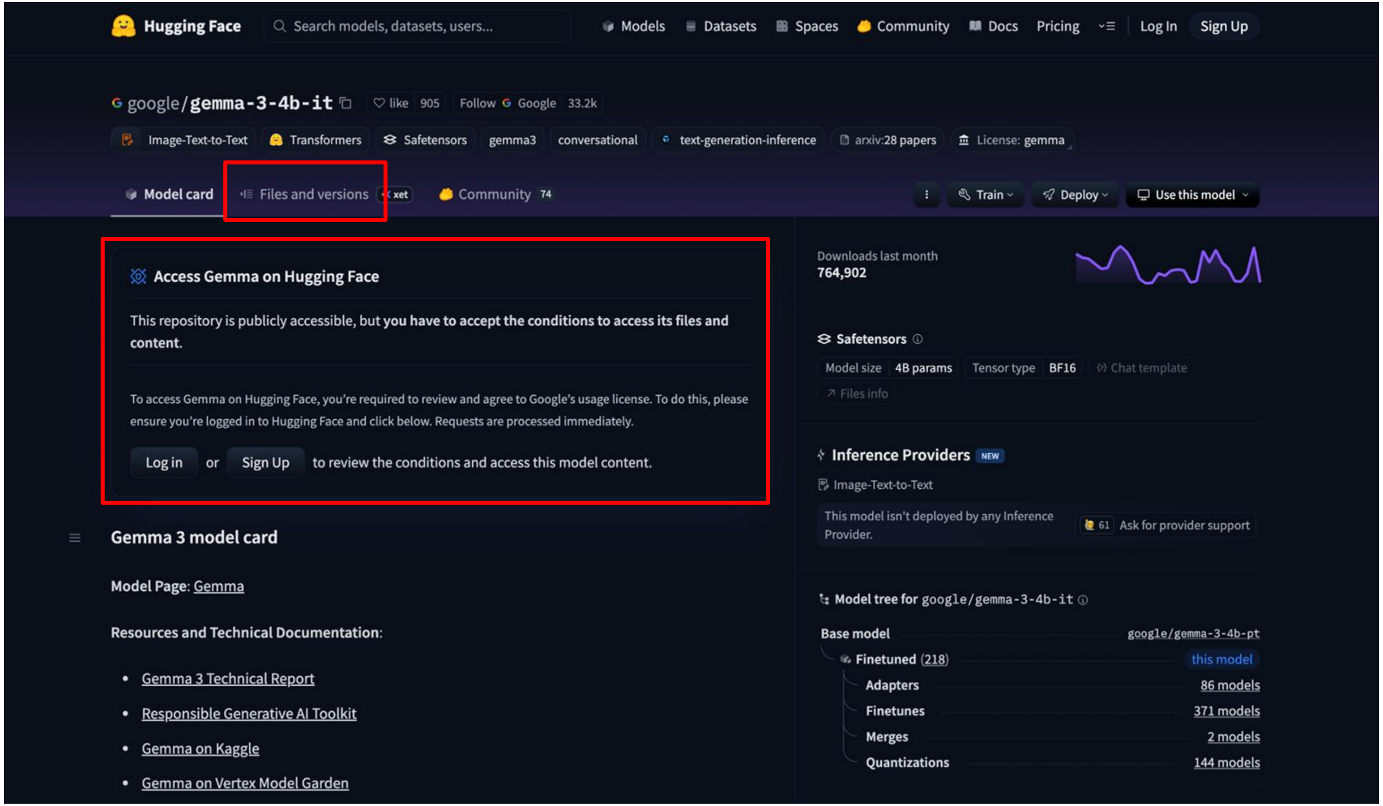
    - Result with RAG - 20%

    - Compare the result (w/o RAG) - 20%

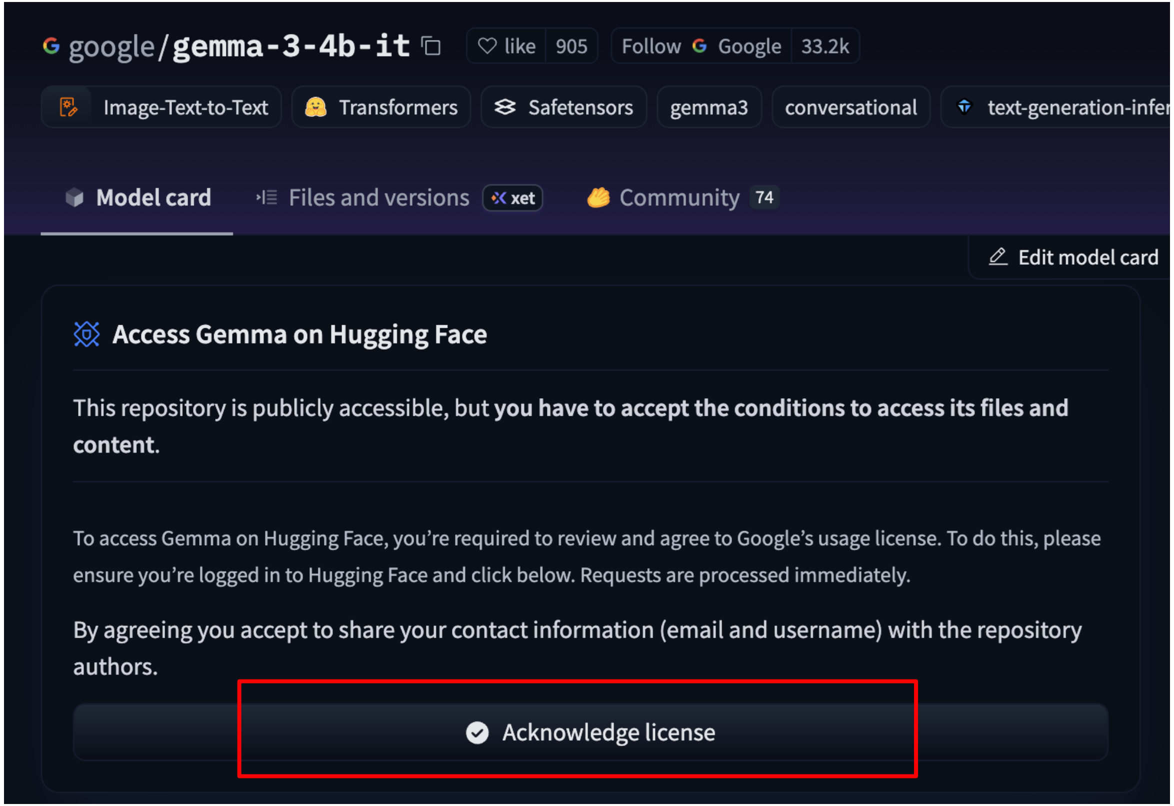
    - Bonus: Different model comparison - 10%

    - Bonus: Describe your own RAG pipeline / Compare with sample notebook - 10%

Access model weight



Access model weight



Access request

Google

Gemma Access Request

Choose Account for Consent

2 Gemma Consent Form

FIRST NAME * *

Enter First Name

LAST NAME * *

Enter Last Name

VERIFIED EMAIL

a0966096979@gmail.com

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Section 1

DEFINITIONS

1.1 Definitions

Run on colab

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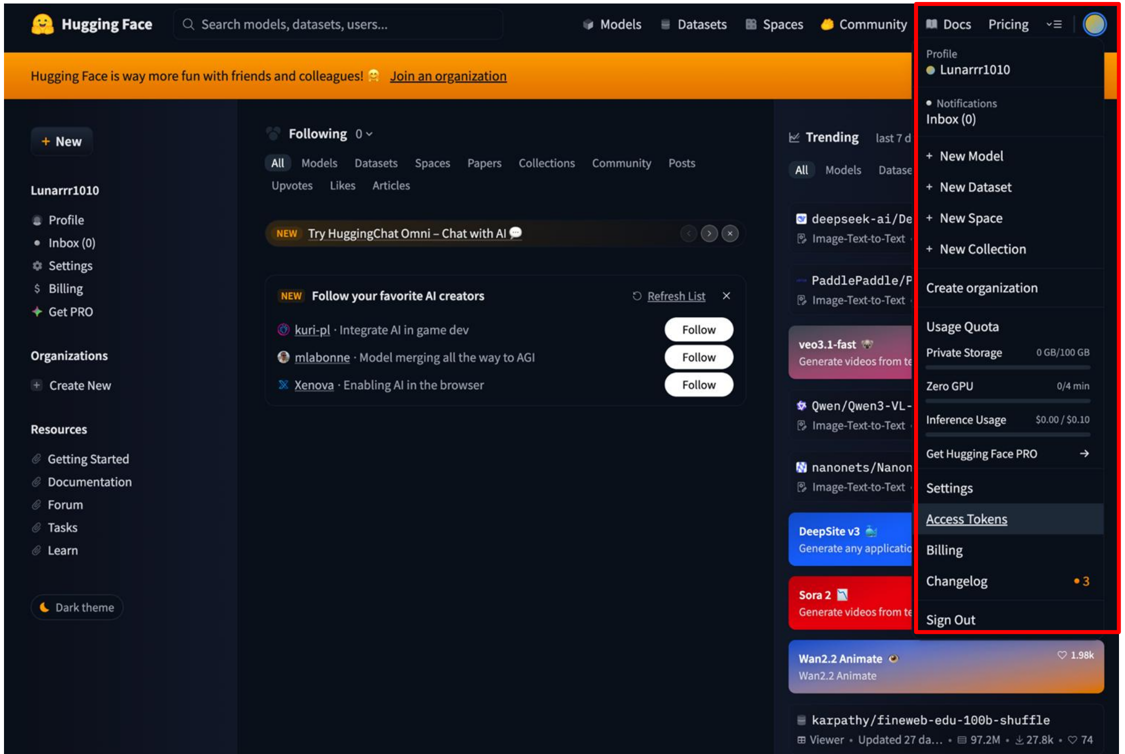
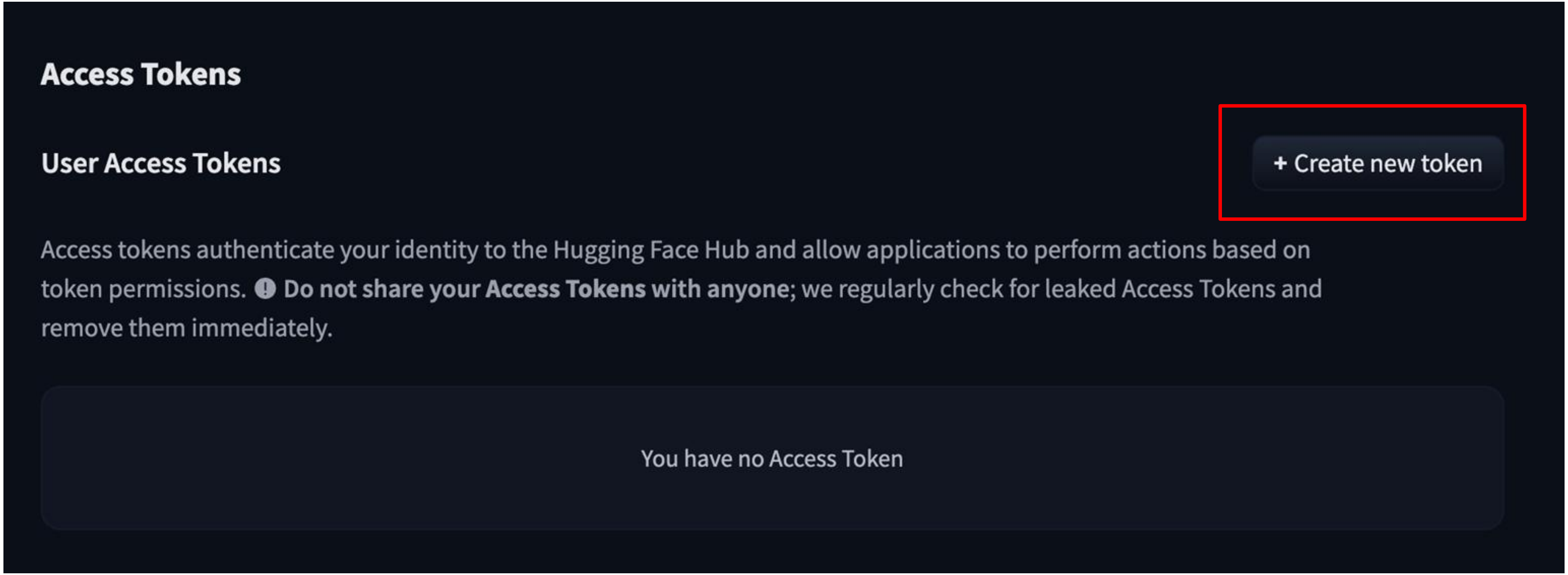
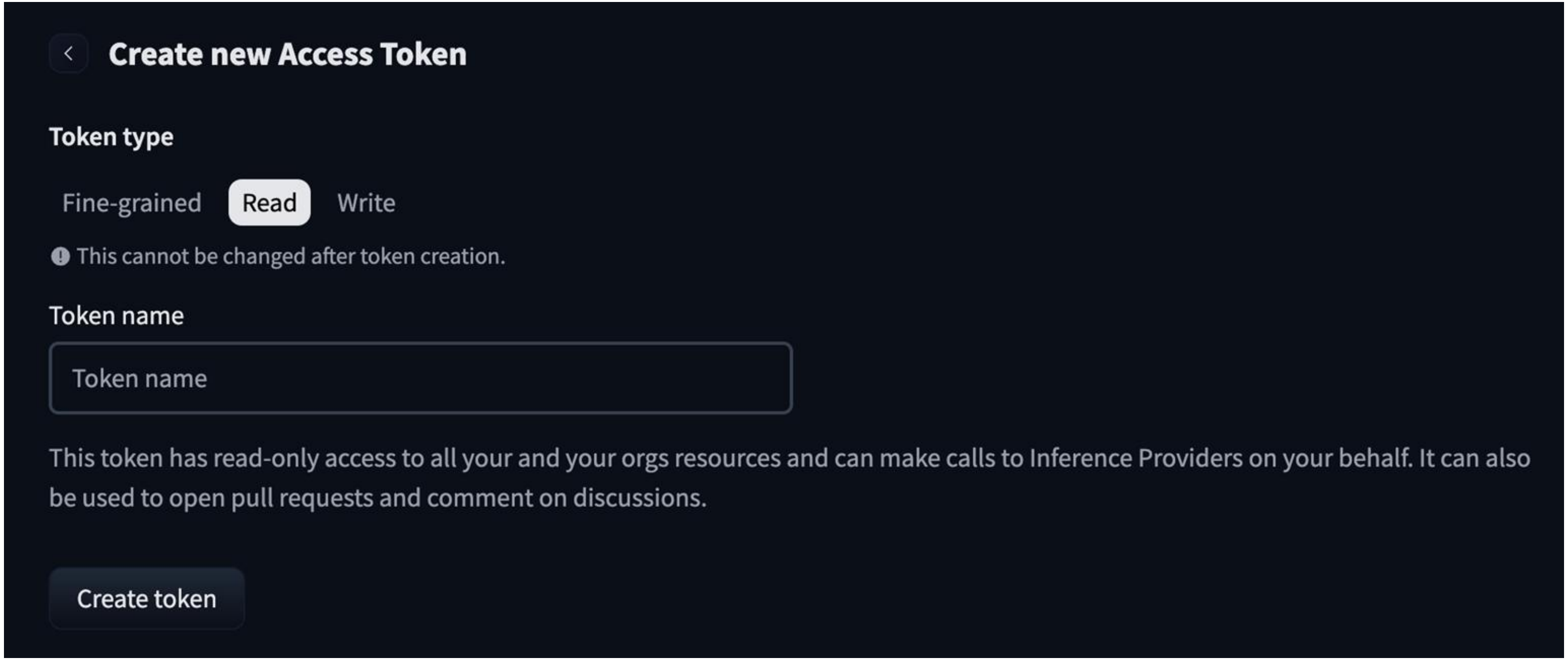
!huggingface-cli login --token hf_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```




Access Tokens
User Access Tokens
+ Create new token
Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions. Do not share your Access Tokens with anyone; we regularly check for leaked Access Tokens and remove them immediately.
You have no Access Token


Create new Access Token
Token type
Fine-grained
Read
Write
This cannot be changed after token creation.
Token name
Token name
This token has read-only access to all your and your orgs resources and can make calls to Inference Providers on your behalf. It can also be used to open pull requests and comment on discussions.
Create token

Referencies

[Sentence transformers]

[LLM Models]